Transfer Evaluation of GRPO vs. turn-PPO on the ASearcher Suite: Qwen3-8B-Base step-75 checkpoints, 10 benchmarks, three scorers

CA-benchmark rung-3 / 8B thread

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Abstract

We evaluate the two finished 8B training arms — token-level GRPO and turn-level PPO (`gae_turn`) — at step 75 on the full ASearcher test suite: 7,152 questions over 10 benchmarks, greedy decoding, the training environment's own wiki-18/e5 retrieval, scored by strict exact match (the training reward), substring EM, and a gpt-oss-120b judge with a frozen prompt. turn-PPO transfers ahead of GRPO on the wiki-answerable slice (macro judge **0.550 vs. 0.535**; strict EM 0.405 vs. 0.395), directionally consistent with the training validation ordering (0.561 vs. 0.520) but with a much smaller margin — and it does so at **2–4× fewer turns per question**. Strict EM understates true accuracy by ≈0.13–0.14 with 99.7% one-directional disagreement. The live-web trio (GAIA / frames / xbench-deepsearch) floors for both arms, measuring the wiki-18 corpus gap rather than the credit-assignment method.

Setup

Checkpoints: `token_grpo` / `turn_ppo_b0 _qwen3-8b-base_32turn_fast75_s0` at step 75 (unfiltered ASearcher Base-35k, 32-turn horizon, 16k prompt / 1024 response, top-5 wiki-18/e5 retrieval, fast config: partial rollout 8/4 + dynamic token batching + chunked prefill). Evaluation: `val_only` greedy rollouts in the training environment; full trajectory dumps retained. Scoring pipeline `scripts/asearcher/judge_rollouts.py`: dump → trajectory fold → gold join → concurrent judging (gpt-oss-120b, TP4, mxfp4; frozen prompt `asearcher_mbe_v1`) → per-benchmark aggregation. The 7 *wiki-answerable* benchmarks (NQ, TriviaQA, PopQA, HotpotQA, 2Wiki-MultihopQA, Musique, Bamboogle; 6,115 questions) are reported separately from the 3 *live-web* benchmarks (GAIA, frames, xbench-deepsearch; 1,027 questions) that require open-web retrieval our offline corpus cannot serve.

1 Headline scores

Wiki-answerable (7 benchmarks, macro)		
scorer	GRPO	turn-PPO
judge	0.535	**0.550**
strict EM	0.395	**0.405**
sub-EM	0.446	**0.461**

Live-web (3 benchmarks, macro)		
scorer	GRPO	turn-PPO
judge	0.097	**0.126**
strict EM	0.061	**0.078**
sub-EM	0.066	**0.087**

Table 1: Macro scores at step 75. turn-PPO leads every scorer on both slices; the wiki-answerable margin (judge +0.015) is far smaller than the training-val margin (+0.041 strict EM). Live-web numbers measure the corpus gap, not the method.

2 Per-benchmark results

3 Search efficiency

GRPO runs essentially every question to near the 32-turn cap (16.5–18.1 average turns, uniformly — including single-hop questions answerable with one search), consistent with its training-side turn re-inflation (14.6→17.0 turns over steps 50–75) and per-turn response collapse (54 tokens/turn). turn-PPO modulates its budget with difficulty: the full distribution (Fig. 4) is **bimodal** — 44% of its trajectories answer within ∼2 turns (single-hop

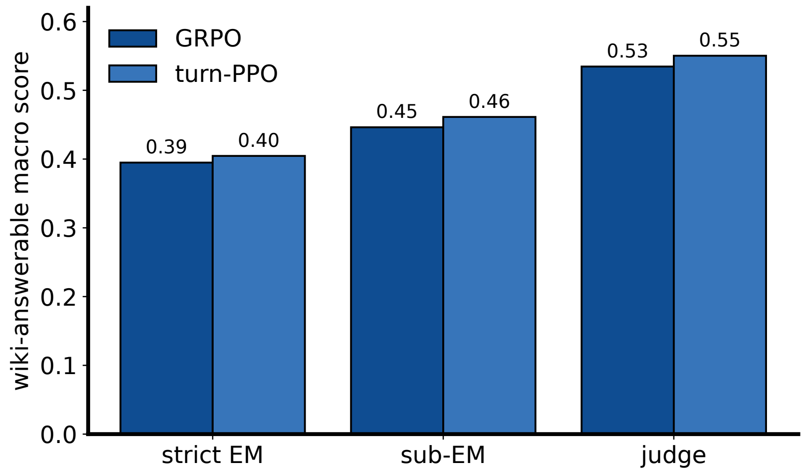

Figure 1: Wiki-answerable macro score under the three scorers. The EM→judge gap (≈0.13–0.14 for both arms) is the formatting/alias slack of strict EM — the metric the environment actually rewards during training.

		judge		strict EM		avg turns	
benchmark	n	GRPO	tPPO	GRPO	tPPO	GRPO	tPPO
NQ	999	**0.616**	0.614	0.428	**0.433**	18.1	5.5
TriviaQA	1000	**0.776**	0.772	**0.577**	0.572	18.0	4.3
PopQA	992	**0.505**	0.501	**0.446**	0.438	17.5	4.7
Hotpot	1000	0.592	**0.615**	0.401	**0.419**	17.2	8.5
2Wiki	1000	0.480	**0.520**	0.389	**0.407**	16.5	12.3
Musique	999	0.245	**0.286**	0.147	**0.179**	17.6	11.4
Bamb.	125	0.528	**0.544**	0.376	**0.384**	17.8	10.8
GAIA *	103	0.049	**0.087**	0.049	**0.078**	16.9	10.2
frames *	824	**0.233**	0.231	**0.135**	0.125	17.1	11.1
xbench *	100	0.010	**0.060**	0.000	**0.030**	7.7	9.9

Table 2: Per-benchmark scores (* = live-web). Single-hop (NQ/TriviaQA/PopQA) is a dead heat (±0.01); turn-PPO wins every multi-hop benchmark (+0.02–0.04 judge).

questions resolved immediately) with a second mode at ∼13 turns for multi-hop — while GRPO is unimodal at 16–19 turns for *everything* (medians 12 vs. 17, means 8.3 vs. 17.3). At equal-or-better accuracy this is a **2–4× cheaper inference-time policy**.

4 Training dynamics

The strategy split emerges during training, not at eval time (Fig. 5). Both arms' temperature-1 train success climbs 0.2→0.5 nearly in lockstep — the training reward barely distinguishes them — while their *policies* diverge: both inflate to ∼570 response tokens/turn by step 5, then GRPO collapses to ∼54 tokens/turn (terse queries, many turns) while turn-PPO stabilizes at ∼145 (fuller per-turn reasoning, fewer turns). The eval-time differences of the preceding sections are the downstream signature of this divergence.

5 Scorer calibration

6 Caveats

- **Single seed.** One run per arm; the wiki-answerable judge gap (+0.015) is small. Seed replication is queued before any headline claim.

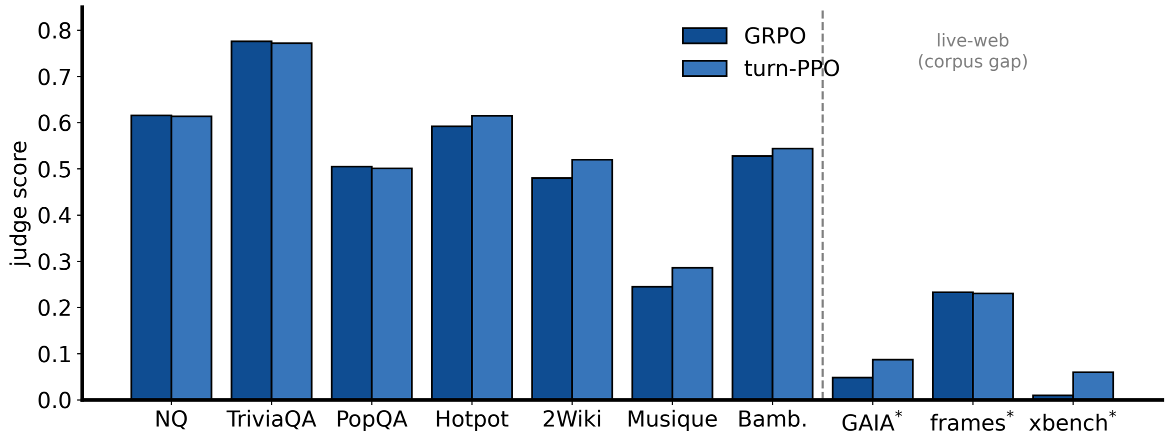

Figure 2: Judge score by benchmark.

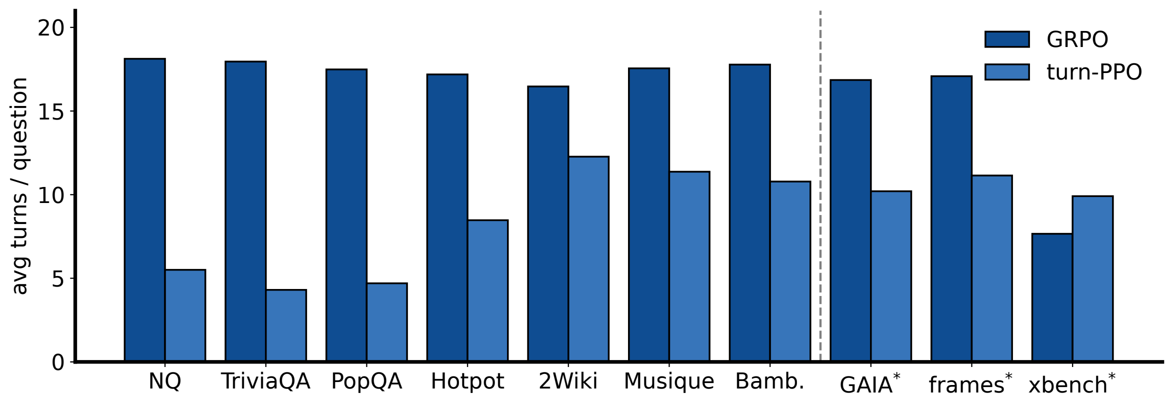

Figure 3: Average turns per question. Dotted line = 32-turn cap.

- **Unfiltered training data (parametric-recall confound).** Both arms trained without the closed-book filter. Measured on a 944-query sample of the training pool (early-training 8B, temperature 1, 5 attempts each; Fig. 6): the near-base policy solves 51.6% of queries at least once, and **54.8% of those solved queries have at least one zero-search win** (27.3% are solved *only* by memory) — 43.9% of all winning trajectories used no search at all. The training reward therefore partly pays for parametric recall rather than search skill. The trained checkpoints nearly always search at eval time (zero-search trajectories: GRPO 0.0%, turn-PPO 1.6%), but this does not prove retrieval *dependence*, and the 4B *filtered* campaign inverted the arm ordering. (For scale: the forced closed-book measurement on 4B found 28.8% of all queries parametric-answerable; the 8B in-env 28.3% is a lower bound on the same quantity, so the true 8B rate is likely higher. The forced 8B closed-book pass is queued.)
- **Live-web $\neq$ method comparison.** GAIA/frames/xbench need the open web; our retriever serves a 2018 wiki snapshot.
- An earlier (2026-08-04 morning) claim that transfer *reverses* the training-val ordering was based on a mis-computed turn-PPO macro (0.355) and is superseded by the full-dump judged numbers here.
- **Pending.** The three in-flight arms (`hcapo_ans`, `token_ppo`, `gigpo`) receive the identical eval + judge pass at step 75, extending this to the full five-arm estimator comparison.

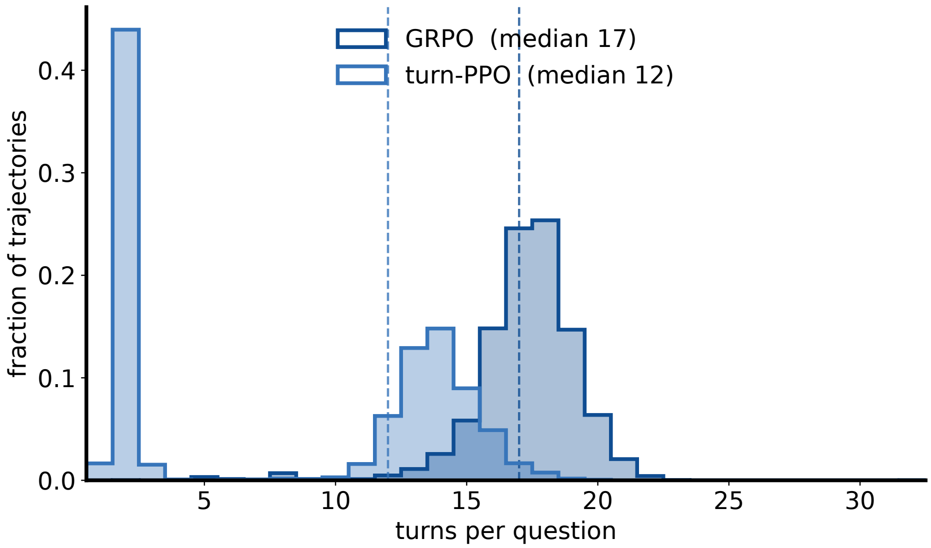

Figure 4: Distribution of turns per question over all 7,152 eval trajectories. turn-PPO is bimodal (immediate answers on single-hop, ∼13 turns on multi-hop); GRPO spends 16–19 turns regardless of difficulty. Dashed lines = medians. Neither arm hits the 32-turn cap.

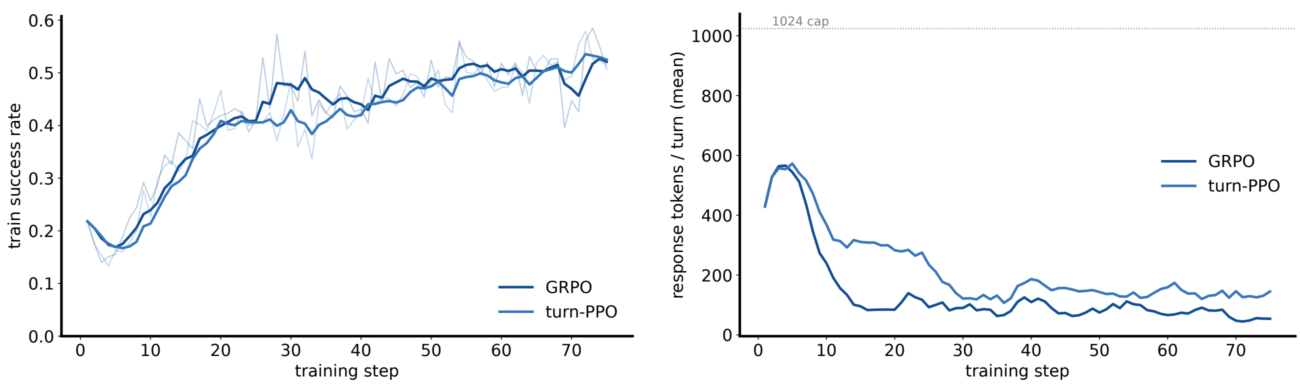

Figure 5: Training dynamics over the 75-step runs. Left: temperature-1 train success (faint = raw per step, bold = EMA). Right: mean generated tokens per turn; dotted line = 1,024-token cap.

arm	judge–EM agr.	judge✓EM×	judge× EM✓	EM vs. env reward	no-answer frac
GRPO	0.870	930	3	1.000	0.080
turn-PPO	0.863	974	3	1.000	**0.005**

Table 3: Scorer agreement over 7,152 trajectories per arm. Disagreement is 99.7% one-directional (judge-correct, EM-wrong): aliases, formatting, paraphrases. *EM vs. env reward* = 1.000 confirms the pipeline reproduces exactly the reward the environment assigned. GRPO leaves 8.0% of questions with no <answer> at all (loops to the cap); turn-PPO 0.5%. Judge failures: 0. Manual smoke calibration: 0.983 agreement on 60 hand-checked items.

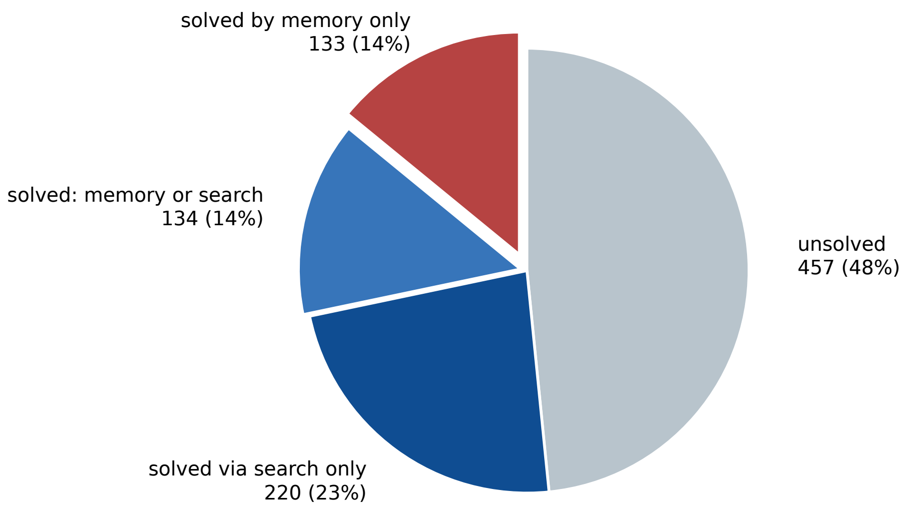

Figure 6: The parametric-recall confound in the unfiltered training data: 944 training queries $\times$ 5 attempts by the early-training 8B (temperature 1). Roughly half the queries it can solve at all admit a zero-search (memory-only) win; for 14% of all queries, memory was the *only* successful route. These queries pay training reward regardless of search skill.